PV-Finder: primary vertex reconstruction and track association with neural networks

Paper deck: configuration, definitions, answers and figures

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This Deck

For the colleague finishing the paper draft (`PVFinderPaper_11_11_2026.pdf`).

1. Frozen configuration, one table per regime
2. Datasets and which file is used for what
3. Metric definitions, with the code that computes each one
4. Answers to the 29 questions in the draft, with page and line
5. Answers to Mattia's own questions
6. Results: every paper figure and number, main and backup
7. Proposed internal-note edits

How to read it

- Every number is a macro in `numbers.tex`.
- TBD marks a number from the final recompute that is not in yet. Its source file and key are in the comment next to the macro.
- A framed red box marks a figure not produced yet, with its expected path.
- **PAPER MAIN** figures are proposed for the paper; **BACKUP / NOTE** figures go to the note or the backup.
- PV-Finder and AMVF are always compared on the same events with the same code.

1

Frozen Configuration

One configuration per regime, used for every result

HL-LHC Configuration ($0 \le \mu \le 200$)

Density model	s8: U-Net v2, 80 channels, 1 latent channel MLP $7 \to 5 \times 128 \to 1000$, 439,593 parameters
Checkpoint	`hllhc_alleta_v6_s8_1p1ep_phase2_step_450000`
Peak amplitude τ	0.01
Minimum width	3 bins (0.04 mm each)
Minimum integral	0.184
Minimum height	0.0184
Position	local centroid, ± 3 bins about the maximum
GNN	full-μ v2, epoch 527, 39,521 parameters
Graph	$k = 20$ nearest peaks per track
Score cut t	0.97 (top-1 edge per track kept if score $\ge t$)
Output vertices	candidates with ≥ 1 assigned track
Matching window	0.17 mm (all vertex results, both algorithms)
Hard-scatter window	0.1 mm
$\sigma_{\mathrm{vtx-vtx}}$ fit	counted peaks, 300 bins over ± 6 mm

- One line in code: `get_regime("hllhc")` in `src/pv_finder/utils/working_points.py`.
- The integral and height cuts are the v6 cuts (0.30 / 0.03) scaled to give s8 the v6 fake rate.
- The $\sigma_{\mathrm{vtx-vtx}}$ fit uses the same peaks as the counts. There is no second threshold.
- Clean requires $\ge 70\%$ of a vertex's tracks from one truth vertex.
- Single-vertex residuals use an optimal one-to-one assignment within 1 mm. This is a measurement radius, not a matching window.

Low Pile-up Configuration (Run 2 MC, Run 2 and Run 3 Data)

Density model	PVF-e400: U-Net v1, 64 channels, 1 latent channel MLP $7 \to 5 \times 100 \to 1000$, 358,713 parameters
Checkpoint	`e2e_mlpHist50_e2e400_1latent_mse_phase2_epoch_400`
Training	50 epochs MLP to histogram, 400 end to end; no KDE
Peak amplitude τ	0.01
Minimum width	3 bins
Minimum integral	0.50
Minimum height	none
Position	local centroid, ± 3 bins
GNN	`ttva_gat_mu60_v2_aug epoch 115`
Score cut t	0.97 (same as HL-LHC)
Output vertices	candidates with ≥ 1 assigned track
Matching window	0.17 mm
Hard-scatter window	0.1 mm
$\sigma_{\text{vtx}-\text{vtx}}$ fit	counted peaks, 300 bins over ± 6 mm

- `get_regime("lowpu")`, same file.
- One density model for every low pile-up result. The ep300 reproduction model and the two-stage KDE model are no longer used.
- The window is the HL-LHC value. At low pile-up it is TBD times the PV-Finder 68% residual of Run 2 MC.
- $t = 0.97$: no other t is better in Clean/truth, HS identification and track F_1 together on Run 2 MC.
- There is no Run 3 MC. The simulation is called Run 2 MC.

What Changed with Respect to the 11 November Draft

Item	Draft of 11 November	Final
HL-LHC density model	v6 (280 channels, 4 latent, 3.55 M parameters)	s8 (80 channels, 1 latent, 439,593)
HL-LHC peak cuts	integral 0.30, height 0.03	0.184, 0.0184
$\sigma_{\mathrm{vtx-vtx}}$ threshold	separate peak finding at integral 0.50	the counted peaks
Matching window	0.1694 mm; 0.5 mm in the chain; fitted $\sigma_{\mathrm{vtx-vtx}}$ at low μ	0.17 mm everywhere
HL-LHC GNN	v4 (PU200 only), epoch 107	full-μ v2, epoch 527
GNN score cut	0.98 (categories), 0.95 (HS)	0.97 for every use
Low pile-up density model	ep300 (KDE-initialised), PVF-e400, KDE two-stage	PVF-e400 only
Low pile-up GNN peaks	integral 0.2, full-region mean	integral 0.50, local centroid
Low pile-up t	0.99, 0.95 (F_1), 0.5 (HS), 0.98 (Run 3)	0.97 for every use
HS selection	plain $\sum p_{\mathrm{T}}^2$	Athena 24.0 rule, same code for both
HS position	histogram peak	3D adaptive fit with beam spot
Categories	AMVF TTVA links vs GNN tracks	same, plus weighted cross-checks

2

Datasets

Training, validation and test samples

HL-LHC Files

File (mc21 14 TeV $t\bar{t}$, e8481_s4494, ITk)	Events	Pile-up	Role
SingleLep 601229 r16438	99,800	190–210, $\langle\mu\rangle = 200.0$	density training pool
SingleLep 601229 r16633	99,800	190–210	density training pool
all-had 601237 r16633, part 1	1,421,400	190–210	density training pool
all-had 601237 r16633, part 2	1,498,600	190–210	density training pool
all-had 601237 r16633, part 3	1,200,000	190–210	density training pool
all-had 601237 r16633, part 4	854,000	190–210	density training pool
PU200 pool	**5,173,600**	integers 190–210	
SingleLep 601229 r16443	100,000	flat 0–200, $\langle\mu\rangle = 99.6$	entries 0–24,999: **test** of every result entries $\geq 25{,}000$: GNN v2 training and validation
SingleLep 601229 r16638	99,800	flat 0–200, $\langle\mu\rangle = 99.6$	GNN v2 training (89,820) second held-out file for the density model

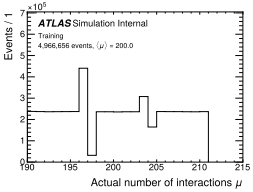
- The density model never saw r16443 or r16638. The GNN never saw r16443 entries 0–24,999.
- Track selection: ITk paper Table 6. In-time pile-up only. Pile-up is `ActualNumOfInt`.
- PU200 = the 1,920 r16443 test events with $185 \leq \mu \leq 215$ (actual 185–200, $\langle\mu\rangle = 192.5$), 213,738 truth vertices with $n_{\text{trk}} \geq 2$.

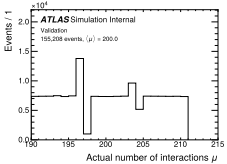
`docs/additional_studies/paper_questions_2026-11-11.md Part 3; outputs/11_11_2026_output/datasets/hllhc_dataset_inventory.json`

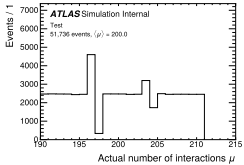
HL-LHC Density Model: Training, Validation and Test

Role	40 mm intervals	Event equiv.
Training (96%)	59,599,872	4,966,656
Validation (3%)	1,862,496	155,208
Test (1%, unused)	620,832	51,736

- Seed-42 random permutation of the intervals of the PU200 pool. The roles therefore share its pile-up shape: uniform over 190–210, with $\mu = 196$ doubled and $\mu = 197$ nearly empty (production feature).
- Results never use the test intervals. They use the independent files r16443 and r16638.

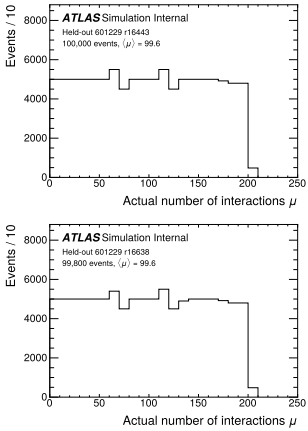






HL-LHC Held-out Files and the GNN v2 Samples

GNN v2 set	Source	Events
Training	r16443 entries $\geq 25{,}000$	60,000
	r16638 (stratified in 10-unit μ bins)	89,820
	total	149,820
Validation (t choice)	r16443 entries $\geq 25{,}000$	7,500
monitor (checkpoint)	subset of validation	1,500
Reserved test	r16443 entries $\geq 25{,}000$	7,500
Reserved test	r16638	9,980
Paper test	r16443 entries 0–24,999	25,000



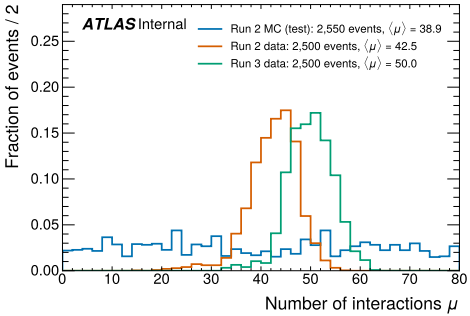
- Graphs use truth-vertex candidates, so no density model enters the GNN training data.
- No r16638 truth-z or HS (x, y, z) fingerprint occurs in r16443.
- Resolution vs μ (before the GNN): 44,999 events of both files, fixed per-bin counts chosen before looking at results.

`outputs/11_11_2026_output/gnn_fullmu_v2/split_manifest.json; docs/evaluation/gnn_full_pileup_training_v2_2026-11-11.md`

Low Pile-up Samples

Sample	Events	Use
Run 2 MC ($t\bar{t}$, μ 0–80)	51,000	
training (70%)	35,700	density model
validation (25%)	12,750	density model
test (5%)	2,550	**every MC result**
non-test (train + val)	48,450	GNN training
Run 2 data (ZeroBias, run 364076)	2,500	data
Run 3 data (`file_3.root`)	2,500	data

- MC test: h5 events 48,450–50,999 (the last 5%), 72,189 truth vertices with $n_{\text{trk}} \geq 2$, $\langle\mu\rangle = 38.9$.
- Run 2 data: entries 0–2,499. Run 3 data: the first 2,500 accepted entries of 0–3,158. Vertex finding and track association use the same events.
- Data have no truth: AMVF vertices ($n_{\text{trk}} \geq 2$, beam-spot corrected) are the reference.



Pile-up of the three evaluation samples.

3

Metric Definitions

How each number is computed, and the code that computes it

Before the GNN: Efficiency, Fakes, Splits

Quantity	Definition	Code
Efficiency (Eq. 1)	truth vertices ($n_{\mathrm{trk}} \geq 2$) with ≥ 1 reco vertex within 0.17 mm, over all truth vertices; ratio of sums over events. Equals clean + merged truth of the closest-first matcher (merged credit), event by event.	`compare_res_reco` in `efficiency_res_optimized_atlas.py`; identity test `tests/test_amvf_fair_matching.py`
One-to-one efficiency	truth vertices that win a dedicated reco vertex in the closest-first assignment	same
Fake fraction	reco vertices with no truth vertex within 0.17 mm, over all reco vertices	same (`fake`)
Fakes per event	the same count over the number of events	same
Split	reco vertex with a truth vertex in its window that a closer reco vertex already took	same (`split`)
HS reconstruction	events with a reco vertex within 0.1 mm of the true HS (z only, ITk Sec. 5.5), over all events	`gnn/evaluation/hs_metrics.py`

Truth: `TruthVertex` with ≥ 2 selected tracks. AMVF: stored vertices with $n_{\mathrm{trk}} \geq 2$. Errors: event bootstrap (2,000 replicas); PV-Finder minus AMVF is paired on the same resampled events.

Before the GNN: Resolution and the Window

Quantity	Definition	Code
$\sigma_{\text{vtx}-\text{vtx}}$	all within-event pairs of reco vertices; Δz in 300 bins over ± 6 mm; fit $a/(1+e^{b(r-\lvert x\rvert)})+c$; $\sigma_{\text{vtx}-\text{vtx}} = \lvert r\rvert$. Same peaks as the counts. Quoted at PU200, and per μ bin only for $\mu \geq 100$.	`fit_dz` in `run_eval_pvf_run3.py`; `amvf_fair_comparison/scan.py`
68% residual	68th percentile of $\lvert z_{\text{reco}} - z_{\text{true}}\rvert$ after the optimal one-to-one assignment within 1 mm (most matches, then least total distance). The resolution versus μ.	`matched_residuals_optimal`, `amvf_fair_comparison/matching.py`
Window	$3\times$ the larger 68% residual of the two algorithms at PU200, rounded: $3 \times 56.45\ \mu\text{m} = 0.1694$ mm $\rightarrow$ 0.17 mm	`working_points.MATCH_WINDOW_MM`
Hard-scatter window	0.1 mm, ITk performance paper	`working_points.HS_MATCH_WINDOW_MM`

After the GNN: Vertex Categories

Quantity	Definition	Code
Assignment	each track keeps its top-1 GNN edge if the score is $\geq t = 0.97$; candidates with no track are dropped from the output	`gnn/evaluation/chain_inference.py`
Track lists	PV-Finder: GNN-assigned tracks. AMVF: its TTVA links (`Nonprompt_Hard_MaxWeight`, `RecoVertex_assocTracks`). One rule for both.	
Clean, Merged, Fake	per vertex, the plurality truth vertex of its tracks. Fake if the plurality is unlabelled tracks; Clean if its share is $\geq 70\%$; Merged otherwise	`classify_assignments`, `gnn/evaluation/classification.py`
Split	among vertices with the same plurality truth vertex, all but the largest $\sum p_{\mathrm{T}}^2$	same
Per event	Clean, Merged, Split, Fake vertices per event (bar plot at PU200, versus μ)	`hs_metrics.classify_vertices`
Clean/truth	Clean vertices over truth vertices with $n_{\mathrm{trk}} \geq 2$	`chain_t_summary`
Cross-checks	AMVF with its fit tracks and fit weights; PV-Finder with GNN scores as weights	`weighted_classification.py`
Bounds	finder cap: peaks matched one to one within 0.17 mm. Truth-guided: each labelled track to the peak nearest its own truth vertex	`hs_metrics`, `chain_gap_decomposition.py`

Errors: event bootstrap (200 replicas), paired. Every category count is cross-checked by independent code.

After the GNN: Hard Scatter

Quantity	Definition	Code
HS selection	score = $\sum p_{\mathrm{T}}^2$ over a vertex's tracks with compatibility $\chi^2 \leq 18.42$ (Athena 24.0 `SumPt2Sorting`); selected vertex = argmax. Reconstructed and selected: the selected vertex is the HS match (most HS tracks) and lies within 0.1 mm of the true HS. AMVF reference: its stored primary.	`gnn/evaluation/pv_selection.py`; `hs_selection_identical_rule.py`
HS resolution	events where both algorithms have a vertex within 0.1 mm of the true HS; $z_{\mathrm{reco}} - z_{\mathrm{HS}}$ of the closest; width = iterative RMS within ± 3 RMS until stable (Athena `iterRMS_convergence`). PV-Finder position: 3D adaptive fit with beam-spot constraint of the GNN-assigned tracks.	`itk_resolution.iterative_rms`; `hs_position_estimators.adaptive_beamline_fit`
Clean-HS fraction	events with a Clean vertex whose plurality truth is the HS	`hs_metrics`
HS-track fraction	HS tracks on the HS-matched vertex over all HS tracks	same
Local density	truth vertices within ± 2 mm of the true HS, HS excluded, over 4 mm (all truth vertices, ITk definition)	`itk_resolution.local_pu_density`

After the GNN: Track Association

Quantity	Definition	Code
Correct track	assigned to a vertex whose plurality truth vertex is the track's own	`chain_t_metrics`
Track efficiency	correct tracks / truth-labelled tracks; unassigned tracks stay in the denominator	same
Track purity	correct tracks / assigned tracks with a truth label	same
Track F_1	$2\epsilon p/(\epsilon + p)$	same
Wrong rate	$1-$ purity	same
Unassigned fraction	$1-$ (assigned labelled tracks / labelled tracks); efficiency = purity $\times(1-$ unassigned fraction$)$	`hllhc_final_chain_extras.py`

Tracks without a truth label enter neither efficiency nor purity, only the assigned fraction.

4

The 29 Questions

Answers with page and line of the 11 November draft

Q1. Do the 40 mm Intervals Overlap?

Draft p. 7, l. 190 *Do the intervals overlap slightly at their boundaries? If so, by how much, and how is a track in an overlap region assigned?*

- **No overlap.** Interval k covers $[-240+40k,\ -240+40(k+1))$ mm, $k=0,\ldots,11$.
- Each track goes to the one interval that contains its z_0; the lower edge is included.
- Tracks outside $[-240, 240)$ mm are dropped. The ntuple selection already requires $|z_0| \le 200$ mm.
- One sentence worth adding: the target histogram is built on the full 12,000-bin grid and then cut into twelve pieces, so a truth vertex near a boundary can put target content into the neighbouring interval, which holds none of its tracks.

`src/pv_finder/data/root_to_h5.py:34-36, 67-104, 108-140`

Q2. Is Every Element of Fig. 3 Right, and Does It Cover Run 2/3?

Draft p. 8, Fig. 3 caption

Corrections for the final HL-LHC configuration

- **Latent map: $\mathbf{1 \times 1000}$**, not 4×1000. The frozen model s8 has one latent channel (v6 had four).
- **Inputs**: seven features per track, d_0, z_0, $\sigma(d_0)$, $\sigma(z_0)$, $\mathrm{cov}(d_0, z_0)$, interval start, interval end. The helix parameters $(\phi, \theta, q/p)$ of l. 183 do not enter the density network; θ, ϕ, p_T enter only the GNN.
- **GNN update order**: track-to-vertex and vertex-to-track updates run in parallel from the previous layer (one `HeteroConv`), not vertices first (l. 307–308).

Run 2/3 differences

- U-Net v1 instead of v2 (Q10).
- 64 channels instead of 80.
- Fully connected track-vertex graphs instead of $k = 20$.

Final: one figure covers both regimes if the caption lists these three differences. The latent map is 1×1000 in both.

`root_to_h5.py:37; src/gnn/models/ttva_gat.py:67-121; working_points.py`

Q3. Target Widths Come from AMVF

Draft p. 9, l. 205 *The current target-width model is derived from AMVF vertex resolutions. Has anything been done to remove this dependence?*

- **Not for this paper.** The final models are trained on the AMVF-derived widths.
- Width model: $\sigma(n_{\mathrm{trk}}) = A\, n_{\mathrm{trk}}^{-B} + C$, floor 2 μm; truth vertices with fewer than two tracks get one bin (0.04 mm).
- HL-LHC preset `hllhc_alleta`: $A = 0.14318$ mm, $B = 0.362$, $C = -0.01797$ mm, a background-corrected fit of AMVF-truth residuals (Q29).
- Only the widths come from AMVF. The target positions are the truth vertex positions.
- What removing it takes: a preset fitted on the track-only precision (inverse-variance z_0 of the truth tracks, already computed as a diagnostic), a rebuild of the 5,173,600-event pool and a retrain (s8: 13.7 h on one A100).

`root_to_h5.py:82-88; src/pv_finder/data/resolution_presets.py`

Q4. U-Net v2 Details

Draft p. 9, l. 223

Yes, the Appendix E design, with 80 channels in the frozen model (s8):

- **Encoder**: ConvBNReLU($1 \to 80$, kernel 15) at 1000 bins; residual block (kernel 5) at 1000; max-pool 2, residual block at 500; max-pool 2, residual block at 250 (bottleneck, factor 4).
- **Decoder**, twice: nearest-neighbour upsample $\times 2$ + Conv(kernel 3) + BN + ReLU, additive skip, residual block (kernel 5). $250 \to 500 \to 1000$.
- **Head**: ConvBNReLU($80 \to 80$, k5), ConvBNReLU($80 \to 40$, k3), Conv($40 \to 1$, k1), softplus.
- Residual block: Conv + BN + ReLU + Dropout, plus identity.
- **Normalisation**: BatchNorm after every convolution except the last 1×1. ReLU throughout.
- **Dropout** 0.25 after the ReLU of every convolution block; none in the MLP.
- **MLP**: $7 \to 128 \times 5 \to 1000$, LeakyReLU (0.01), softplus; padded tracks masked, output summed over tracks and scaled by 10^{-3}.
- **Parameters**: 439,593 = MLP 196,072 + U-Net 243,521.

Final: Table B.1 (280 channels) and the 3.55 M parameters describe v6. Replace by 80 channels and 439,593.

`src/pv_finder/models/unet_v2.py; autoencoder_models.py:214-352; configs/vertex_finding/config_hllhc_alleta_v6_small_s8.yml`

Q5. Concatenation or Smoothing? One Batch?

Draft p. 9, l. 232

- **Plain concatenation in z order.** No smoothing, normalisation or boundary treatment. The intervals do not overlap (Q1).
- The twelve intervals run as **one batch of 12**, padded to the widest interval.
- In Athena the input tensor is $(12, 7, N_{\mathrm{trk}})$, also one batch.
- Packed-track inference (CPU timing, M6) runs the MLP only on real tracks and scatters each into its interval. The output equals the padded one to 4×10^{-6}; it is the same model.

`run_eval_pvf_run3.py:195-215; src/pv_finder/models/packed_inference.py`

Q6. Phase 1: U-Net Absent or Frozen?

Draft p. 10, l. 244

- **Absent.** Phase 1 runs only the MLP; its latent output is compared directly with the target histogram.
- The U-Net parameters are also set to `requires_grad = False` in Phase 1 and switched on for Phase 2. They have no effect, because the U-Net is not in the forward pass.
- **Final:** the paper says “absent”.

`hllhc_helpers.py:185-195 (phase1_forward); train_hllhc_e2e.py:81-90, 128`

Q7. Datasets, Checkpoint, Warm-up, Time, Hardware

Draft p. 10, l. 250

- **Sample**: the six PU200 files of Part 2, 5,173,600 events.
- **Split**: seed-42 permutation of 62,083,200 intervals: 59,599,872 / 1,862,496 / 620,832 (96/3/1%). Results use r16443 and r16638 only.
- **Schedule of s8**: Phase 1 one epoch, Phase 2 one epoch (v6: 2 + 2).
- **Checkpoint**: Phase 2 step 450,000 of 465,624 (97%). The run stopped on a host-memory error after the last step, before the end-of-epoch checkpoint; step 450,000 is the last saved state. The learning rate there is 1.3×10^{-6}, at the cosine floor.
- **Learning rate**: Phase 1 constant 10^{-3}. Phase 2 per-step linear warm-up from 10^{-6} to 10^{-4} over 3,000 steps, then per-step cosine decay to 10^{-6}.
- **Other**: Adam $(0.9, 0.999)$, batch 128, gradient-norm clip 1.0, MSE loss.
- **Time and hardware**: one NVIDIA A100-PCIE-40GB. Phase 1 9.1 h, Phase 2 4.6 h (resumed once after a cuDNN initialisation crash on the shared GPU). 13.7 h in total.

`outputs/logs/hllhc_alleta_v6_s8_1p1ep_20261109_211919.log, _20261110_211930.log; build_phase2_scheduler, hllhc_helpers.py:100-160`

Q8. Peak Width, σ_{PV} and Height

Draft p. 10, l. 262

- **Width**: square root of the amplitude-weighted variance of the bin index over the whole above-threshold region (up to a split point), times 0.04 mm.
- $\boldsymbol{\sigma}_{\mathrm{PV}}$: this width, used directly in the first edge feature $(z_0 - z_{\mathrm{PV}})/\sqrt{\sigma(z_0)^2 + \sigma_{\mathrm{PV}}^2}$. Median at PU200: TBD mm.
- **Height**: the maximum bin value of the region, the raw predicted density without normalisation. Second vertex-node feature.
- **Training graphs differ**: truth z, σ_{PV} from the `hllhc_alleta` n_{trk} parameterisation, height from the target recipe. The text near l. 441 (Eq. 4) describes σ_{PV} as the n_{trk} parameterisation; at inference it is the peak width.

`efficiency_res_optimized_atlas.py:16-190; graph_construction.py:105-168; chain_inference.py`

Q9. Final Peak-Finding Configurations; Local Centroid at Low Pile-up?

Draft p. 10, l. 269

	HL-LHC	Low pile-up
Amplitude	0.01	0.01
Minimum width	3 bins	3 bins
Integral	0.184	0.50
Minimum height	0.0184	none
Position	centroid ±3	centroid ±3
$\sigma_{\text{vtx}-\text{vtx}}$ peaks	counted peaks	counted peaks

- **Yes**, the local centroid is now used at low pile-up too. On Run 2 MC it changes the efficiency by TBD points.
- The HL-LHC cuts are those of s8. The draft's 0.30 / 0.03 are the v6 values.
- The note's "0.50 for the HL-LHC resolution fit" described the old code. It is gone: the fit uses the counted peaks.

`working_points.py; lowpu_final_numbers_2026-11-12.md Sec. 4`

Q10. Run 2/3 Density Model

Draft p. 11, l. 281

PVF-e400, every low pile-up result.

- **MLP**: $7 \to 100 \times 5 \to 1000$, LeakyReLU (0.01), softplus, 1 latent channel.
- **U-Net v1**, 64 channels, the original design: encoder ConvBNReLU kernels 25, 7, 5, 5 with max-pool, $1000 \to 500 \to 250 \to 125$ (bottleneck 125); decoder three ConvTranspose(2, 2) + ConvBNReLU(k5) with concatenated skips; output Conv($128 \to 64$, k5), Conv($64 \to 1$, k5), softplus; dropout 0.25 in every convolution block.
- **Parameters**: 358,713 trainable.
- **Inputs**: the same seven track features as HL-LHC.
- **Training**: Phase 1, 50 epochs MLP to histogram; Phase 2, 400 epochs end to end; Adam, constant 10^{-3} in both phases, batch 128, MSE. Checkpoint: last epoch (400).
- **Data**: 51,000 Run 2 MC events, 70 / 25 / 5% (Part 2).
- **No KDE in any step**: two-phase, the same procedure as HL-LHC.
- **Target preset**: Qi Bin's file; upstream constants (preset `run3`) $A = 0.23817$ mm, $B = 0.49491$, $C = -0.00079$ mm.

`configs/vertex_finding/config_mlp_hist_e2e.yml; autoencoder_models.py:801-1000; train_mlp_hist_then_e2e.py`

Q11. GNN Details

Draft p. 12, l. 314

- **Layers**: PyG `GATConv` (original GAT, not GATv2), 4 heads averaged (`concat=False`), edge features in the attention (`edge_dim=3`), no self-loops. Two `HeteroConv` layers.
- **Activations**: LeakyReLU (0.01) after the two linear encoders and the two hidden edge-MLP layers; ReLU after each residual sum; sigmoid score.
- **Normalisation**: none. **Dropout**: none. The stored `dropout=0.25` is not used in `forward`, so the trained model has no dropout.
- **Parameters**: 39,521 (v2 epoch 527).
- $\boldsymbol{d_0}$: as stored in the ntuple. In the HL-LHC simulation the beam spot is at $(0, 0, 0)$ with $\sigma_{xy} = 12\ \mu$m, so "relative to the beam spot" and "relative to the origin" are the same.
- **Tracks**: exactly the tracks of the density network; the chain builds both inputs from the same arrays.
- **Track features**: d_0, z_0, $\sigma(d_0)$, $\sigma(z_0)$, cov, θ, ϕ, $p_{\mathrm{T}}/50$ GeV.

`src/gnn/models/ttva_gat.py:54-137; pu200_chain_graphs.py:189-217; constants.py:10`

Q12. Eq. (1) and the Fake Fraction

Draft p. 14, l. 361

- **Eq. (1) equals the note's efficiency exactly**: truth with ≥ 1 reco vertex within the window = clean + merged truth after the closest-first assignment. Checked event by event on every model and on AMVF. No recalculation of the definition is needed; the numbers are recomputed for the final configuration.
- **Fake fraction** (reco vertices without truth within 0.17 mm, over reco vertices), PU200, 1,920 events:

	Fake / reco [%]	Fakes / event	Splits / event
PV-Finder (s8)	TBD	TBD	TBD
AMVF	20.08	19.65	0.184

`metric_definitions.md Sec. 3.3a; model_freeze_candidates_2026-11-12.md; HLDOC`

Q13. One Window for Everything?

Draft p. 14, l. 367 *We think we agreed on a single window of 0.1 mm for all results. Is that correct?*

- **One window, 0.17 mm**, for every vertex result, both algorithms, both regimes. **0.1 mm for hard-scatter results only**, as in the ITk performance paper.
- Not 0.1 mm for all vertices: the 68% single-vertex residual at PU200 is 55–56 μm for both algorithms, and 0.1 mm is less than twice that.
- Every other window in the draft is replaced: 0.1694 mm, the 0.5 mm one-to-one match of the chain, the fitted 0.416 mm (Run 2 MC) and the fitted data $\sigma_{\mathrm{vtx-vtx}}$ are all 0.17 mm in the final numbers.
- The 1 mm of the single-vertex residuals is the assignment radius used to measure the residual that sets the window. It is not a matching window and must stay wider than it.

`working_points.py; matching_window_audit_2026-11-09.md`

Q14. Hard-Scatter Vertex Resolution

Draft p. 15, l. 392

Include it. Definition (ITk Fig. 28, Athena `iterRMS_convergence`):

- events where both algorithms have a vertex within 0.1 mm of the true HS;
- residual $z_{\text{reco}} - z_{\text{HS}}$ of the closest such vertex;
- width = iterative RMS: restrict to the mean ± 3 RMS and recompute until the selection is stable.
- PV-Finder position: 3D adaptive vertex fit with a beam-spot constraint of the tracks the GNN assigns to the HS candidate at $t = 0.97$.

PU200, TBD events	Resolution [μm]
PV-Finder + GNN	TBD $\pm$ TBD
AMVF	10.24 ± 0.27

In the note, say only how it is computed (M9).

`itk_hs_resolution_2026-11-09.md; CH hs_resolution_um`

Q15. Local Pile-up Density

Draft p. 15, l. 396

- **Include it.** Density as in the ITk paper: truth vertices within ± 2 mm of the true HS, HS excluded, over 4 mm. All truth vertices are counted, from Trey's side ntuple (same events, all truth vertices).
- HS reconstruction at 0.1 mm is at least TBD% in every density bin for both algorithms.
- HS selection (same Athena rule): PV-Finder + GNN leads AMVF by TBD to TBD points in every bin.
- HS resolution: PV-Finder + GNN narrower by TBD to TBD μm in every bin, flat in density.
- Also available (backup): vertex efficiency before and after the GNN versus local density.

`hllhc_supplementary_2026-11-12.md; SUP hs_vs_density.json`

Q16. Track Efficiency and Purity Entering F_1

Draft p. 16, l. 428

- **Correct track**: assigned to a vertex whose plurality truth vertex is the track's own truth vertex.
- **Efficiency** ϵ = correct tracks / all tracks with a truth vertex. Unassigned tracks count in the denominator.
- **Purity** p = correct tracks / assigned tracks that have a truth vertex.
- **Tracks without truth** are excluded from both; they enter only the assigned fraction.
- $F_1 = 2\epsilon p/(\epsilon + p)$.

PU200	Efficiency [%]	Purity [%]	F_1 [%]
PV-Finder + GNN	TBD	TBD	TBD
AMVF	TBD	TBD	TBD

`threshold_scan.py:149-164, 240-252; chain_scan.py:96-130`

Q17. Wrong-Assignment and Unassigned-Track Metrics

Draft p. 16, l. 434

- **Include them** (backup figures in Part 6). With N_{lab} labelled tracks, N_{at} assigned labelled tracks, N_c correct:
  - wrong-assignment rate $= (N_{\text{at}} - N_c)/N_{\text{at}} = 1 - p$: PV-Finder + GNN TBD%, AMVF TBD% (PU200);
  - unassigned fraction $u = 1 - N_{\text{at}}/N_{\text{lab}}$;
  - lost unassigned: labelled tracks left unassigned whose truth vertex has no candidate within 0.17 mm: TBD;
  - unlabelled tracks correctly left unassigned / all unlabelled tracks: TBD.
- **Relation to F_1**: $\epsilon = p\,(1 - u)$. The wrong rate is the complement of the purity; the unassigned fraction is what separates efficiency from purity.

`hllhc_final/chain_extras/README.md; CX`

Q18. Significance-Based Association, Eq. (4)

Draft p. 16, l. 445

- **Eq. (4) confirmed**: $s = |z_0 - z_{\mathrm{PV}}| / \sqrt{\sigma(z_0)^2 + \sigma_{\mathrm{PV}}^2}$, the same combination as the GNN edge feature.
- Run on the same graph: each track's $k = 20$ nearest peaks; the track goes to the smallest-s peak if $s < s_{\max}$. σ_{PV} = peak width.
- The best $s_{\max}$ is chosen on the reported events, which favours the baseline.

PU200	Clean/truth [%]
PV-Finder + GNN	TBD
Eq. (4), best $s_{\max}$	TBD
Eq. (4), $s_{\max} = 3$	TBD
nearest z_0	TBD
$\lvert z_0 - z_{\mathrm{PV}} \rvert < 3\sigma(z_0)$	TBD
AMVF	30.94

`graph_construction.py:128; chain_extras README; CX`

Q19. Why 25,000 Events and the 1,920 PU200 Events?

Draft p. 17, l. 453

- **Compute budget**, not physics. The 25,000 are the first entries of r16443 (100,000 events, flat μ 0–200, about 500 per integer μ).
- The 1,920 are those with $185 \le \mu \le 215$ (actual 185–200, $\langle\mu\rangle = 192.5$). The whole file has 7,680 events in that range.
- **The full chain cannot use more of these files**: r16443 entries $\ge 25{,}000$ and r16638 are GNN v2 training and validation data.
- Before the GNN, more events are used where it helps: r16638 is the second held-out file (1,888 PU200 events) and the resolution versus μ uses 44,999 events of both files.

`paper_questions_2026-11-11.md Q19; gnn_full_pileup_training_v2_2026-11-11.md`

Q20. Chance Matches, Window Scan, Efficiency vs μ, n_{trk}, Separation

Draft p. 17, l. 470

- **Window scan**: show it as an auxiliary figure. PV-Finder minus AMVF is between TBD and TBD points for windows of 0.10–0.20 mm.
- **3 mm-shift chance match** at 0.17 mm: PV-Finder TBD%, AMVF TBD%. Keep it in the note.
- **Efficiency vs μ**: available, proposed for the paper.
- **Efficiency vs truth n_{trk}** (PUB-note style): available before and after the GNN.
- **Efficiency vs separation**: available as efficiency versus Δz to the nearest truth vertex, and versus local density (backup). The merged credit of Eq. (1) credits both vertices of a shared peak, so it stays high at small separation; the one-to-one efficiency falls there.

`h1_vertex_finding/plots; chain_extras/plots`

Q21. $\sigma_{\text{vtx}-\text{vtx}}$ Threshold, AMVF Value and Fit

Draft p. 17, l. 481

- **One threshold.** The fit uses the counted peaks (s8: integral 0.184, height 0.0184). The draft's 0.229 mm came from a second peak finding at integral 0.50; v6 at its counted peaks gives 0.218 mm.
- **Final**: PV-Finder $\sigma_{\text{vtx}-\text{vtx}} = \text{TBD} \pm \text{TBD}$ mm, AMVF 0.305 ± 0.006 mm, same 1,920 events.
- **Fit**: all within-event pairs of vertices ($n_{\text{trk}} \geq 2$ for AMVF), Δz histogram with 300 bins over ± 6 mm (0.04 mm), sigmoid $a/(1 + e^{b(r-|x|)}) + c$, $\sigma_{\text{vtx}-\text{vtx}} = |r|$. Same code for both. The error is the fit error.
- The sigmoid assumes a flat plateau of pile-up pairs. It describes the data at high μ only, which is why $\sigma_{\text{vtx}-\text{vtx}}$ is quoted at PU200 (M10).

`run_eval_pvf_run3.py fit_dz; model_freeze_candidates_2026-11-12.md Sec. 2`

Q22. Merged and Split after the GNN at PU200; Empty Candidates

Draft p. 18, l. 500

PU200, per reco [%]	PVF + GNN	AMVF
Clean	TBD	35.2
Merged	TBD	42.6
Split	TBD	22.1
Fake	TBD	0.07
Clean / truth	TBD	30.94

- Table 4 now has the four categories of Table 5.
- **Remove candidates without tracks from the output**, not only from the rates. A vertex without tracks cannot be written as an xAOD vertex with tracks. TBD candidates are dropped at PU200.
- The paper figure is the number of vertices per event in each category (Part 6).

CH table4; SUP classes

Q23. GNN Thresholds at PU200; One Working Point?

Draft p. 18, l. 514

- **One working point, $t = 0.97$**, for vertex classification, HS selection, HS resolution and track F_1, in both regimes. The 0.98 and 0.95 values of the draft are dropped.
- Chosen on the 7,500 GNN validation events (disjoint from the test set) as the most μ-robust value; details in M4.
- At $t = 0.97$, HS reconstructed and selected at PU200 is TBD% (AMVF stored primary 92.92%).

`gnn_v2_freeze_2026-11-12.md Sec. 5; working_points.py`

Q24. AMVF Run 2 Resolution and Efficiency on the Same Events

Draft p. 20, l. 528 Both are now on the same 2,550 Run 2 MC events with the same code.

	$\sigma_{\text{vtx}-\text{vtx}}$ [mm]	Eff., 0.17 mm [%]	Fake/reco [%]	68% res. [μm]
PV-Finder (PVF-e400)	TBD $\pm$ TBD	TBD $\pm$ TBD	TBD	TBD
AMVF	TBD $\pm$ TBD	TBD	TBD	TBD

- The draft's AMVF 0.793 mm was on 2,500 other events; the PV-Finder 0.416 mm used 60 bins and the ep300 model.

LPDOC R1; VF mc

Q25. Low Pile-up GNN at Integral 0.2, Vertex Finding at 0.50

Draft p. 20, l. 535

- **Final: integral 0.50 for both**, on PVF-e400 peaks with the local centroid.
- The difference was more than the threshold: the draft's GNN numbers (77.43% Clean/truth, $t = 0.99$) came from PVF-e400 at integral 0.2 and the full-region mean; the vertex finding from the ep300 model at 0.50.
- **Recalculated**: graphs rebuilt from the frozen model at 0.50, GNN at $t = 0.97$:
  - Clean/truth TBD% (AMVF TBD%);
  - Clean / Merged / Split / Fake per reco: TBD / TBD / TBD / TBD%.
- Integral 0.50 finds fewer candidates than 0.2, mostly small vertices; that and the lower t explain the change from 77.43%.

LPDOC R2, Sec. 4; GM

Q26. GNN Thresholds in Run 2 Simulation

Draft p. 21, l. 548

- **One $t = 0.97$** for vertex classification, track F_1 and HS identification. The 0.99 / 0.95 / 0.5 of the draft are dropped.
- Rule: the HL-LHC value unless clearly worse at low pile-up. No t beats 0.97 in Clean/truth, HS identification and F_1 together by more than two paired standard errors.
- At $t = 0.97$: Clean/truth TBD%, F_1 TBD% (AMVF TBD%), HS identification TBD% (AMVF TBD%).
- $t = 0.99$ gives Clean/truth TBD%, with a lower F_1; its gain comes from $\mu > 40$ only, and at $\mu < 20$ $t = 0.97$ is better.
- The draft's 98.98% HS identification was obtained at $t = 0.99$, not at the 0.5 the note lists.

LPDOC Sec. 3; L/gnn_mc/t_options.md

Q27. Run 3 Data Association: t, Peak Threshold, Events

Draft p. 22, l. 574

- **Final**: $t = 0.97$, PVF-e400 at integral 0.50 with the local centroid.
- **Events**: the same 2,500 Run 3 events and the same peaks as the vertex finding. The draft's 2,000 events were a random sample of `file_3.root`; only 8 were in the vertex-finding set.
- The draft's model was the two-stage KDE model at $t = 0.98$; both are replaced.

Run 3 data, $t = 0.97$	[%]
Tracks assigned (GNN)	TBD
Agreement with AMVF, 1 mm	TBD
Agreement with AMVF, 0.17 mm	TBD
Leading-vertex agreement	TBD

Agreement is over tracks assigned by both. These are agreement numbers, not efficiencies.

LPDOC R4; R3A

Q28. “integral=0.2” in the Fig. 10 Panel Titles

Draft p. 23, Fig. 10 caption

- **0.5 was used**; the titles were stale. Re-running the old checkpoint on 25 Run 2 MC events reproduces the stored peak lists in 23 of 25 events at 0.5 and in 7 of 25 at 0.2; the other 2 differ by CPU/GPU rounding.
- The label came from the April 2026 plotting code, which carried the then-default value.
- **Final**: all panels are regenerated with PVF-e400 at the final configuration. Every panel states “integral threshold 0.50”; simulation panels say “Simulation Internal”, data panels “Internal”; the Run 3 panels name the simulation Run 2 MC.

`LPDOC Sec. 6; L/summary/figures/fig10*`

Q29. AMVF Dependence; Sample of `hllhc_alleta`; Run 2/3 Parameters

Draft p. 28, l. 652

- **AMVF dependence**: Q3.
- **`hllhc_alleta`** was fitted on the **full-$|\eta|$ sample** ($|\eta| < 4$): 601229 r16633, 99,800 events, 2.31 M truth vertices with $n_{\text{trk}} \geq 2$, background-corrected fit of AMVF-truth residuals. Not the $|\eta| < 2.5$ sample.
- Parameters: $A = 0.14318$ mm, $B = 0.362$, $C = -0.01797$ mm.
- **Run 2/3** (preset `run3`, Qi Bin's file): $A = 0.23817443$ mm, $B = 0.49491396$, $C = -0.000787436$ mm.

`src/pv_finder/data/resolution_presets.py; outputs/07_31_2026_output/amvf_resolution_alleta/`

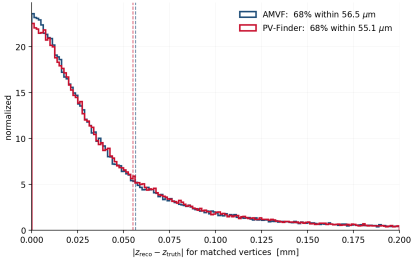
5

Mattia’s Questions

Decisions behind the configuration

M1. Why a 0.17 mm Window

- The window asks whether a vertex was put where the truth vertex is. It is set by how precisely one vertex is placed, the same for both algorithms.
- PU200, 1,920 events: reco and truth vertices assigned one to one within 1 mm (most matches, then least total distance). 68% residual: PV-Finder 55.05 μm, AMVF 56.45 μm.
- Window $= 3\times$ the larger $= 3 \times 56.45\ \mu\text{m} = 0.1694$ mm, rounded to 0.17 mm. Not a Gaussian 3σ: the residuals have tails.
- A greedy assignment would bias the width about 10% low.



Single-vertex residuals at PU200 (v6, 9 November). The s8 version is in Part 6.

Not $\sigma_{\text{vtx}-\text{vtx}}$: it measures how close two vertices can be and still be separated, and differs by 40% between the algorithms. At low pile-up the window is TBD times the PV-Finder residual.

M2. Hard Scatter at 0.1 mm, for ITk Consistency

- The ITk performance paper (arXiv:2412.15090, Sec. 5.5, Fig. 26b) reconstructs the hard-scatter vertex with a 0.1 mm criterion and quotes only the hard scatter. The HS has tens of tracks and a resolution near 10 μm, so 0.1 mm is wide for it.
- The all-vertex population is dominated by low-multiplicity pile-up vertices with a 68% residual of 55–56 μm; for them 0.1 mm would mix position precision into the efficiency. They keep 0.17 mm.
- Every HS number uses 0.1 mm: reconstruction, selection, resolution.
- HS reconstruction at PU200: PV-Finder TBD%, AMVF 99.79%.

Sentence for the paper: “For consistency with ITk performance studies, which quote only the hard-scatter vertex, we also give the hard-scatter results with a 0.1 mm window.”

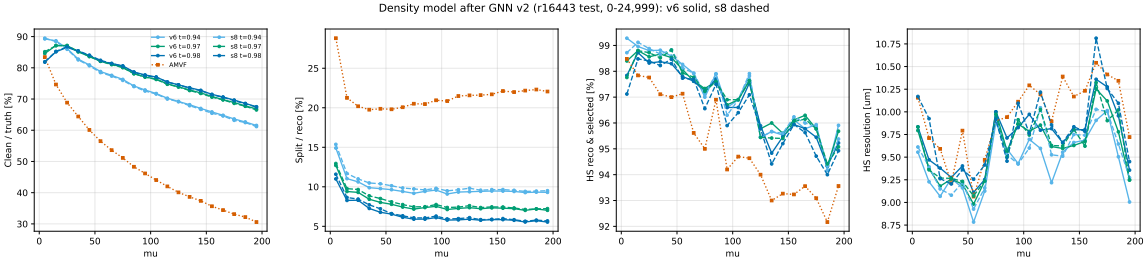
M3. Why s8 Was Frozen

s8 minus v6, same events	
Efficiency at the v6 fake rate (r16443, r16638)	-0.15, -0.08 pt
$\sigma_{\mathrm{vtx-vtx}}$	$+1.2 \pm 1.1$ μm
68% residual	$+0.6$ μm
Clean/truth after the GNN ($t = 0.97$)	-0.14 ± 0.02 pt
HS selection after the GNN	-0.10 ± 0.06 pt
HS resolution after the GNN	$+0.04 \pm 0.04$ μm
CPU speed-up, packed, vs v6 as deployed	9.0× (9 Nov), TBD× (12 Nov)
Parameters	439,593 vs 3,547,993

- Bar: efficiency loss below 0.2 points at the v6 fake rate, and no worse resolution. Only s8 meets it.
- s5 (112 channels) separates close vertices worse: $\sigma_{\mathrm{vtx-vtx}}$ $+9$ to $+12$ μm at 6–7 standard deviations, with either schedule.
- Cuts: v6's 0.30 / 0.03 scaled until the fakes per event match v6; the two-file mean is 0.184 / 0.0184.
- After the GNN the trends with μ are the same for both models.

`model_freeze_candidates_2026-11-12.md; gnn_v2_freeze_2026-11-12.md Sec. 7; cpu_small_model_2026-11-09.md`

M3. s8 and v6 after the GNN



Chain metrics versus μ for both density models with GNN v2, test events (12 November freeze). The trends are the same; the differences are at most a few tenths of a point.

M4. The GNN Score Cut $t = 0.97$

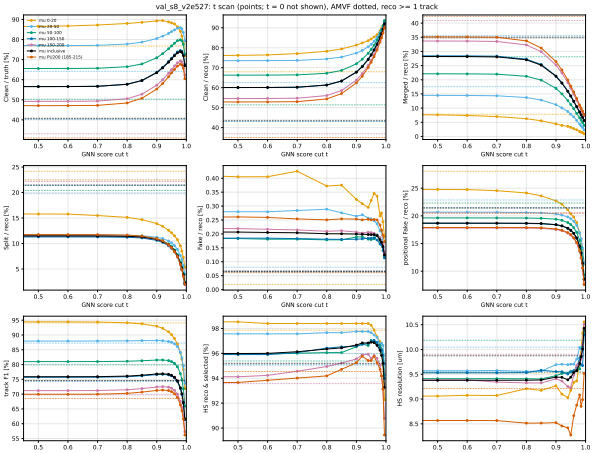
Rule (on validation)	t	Clean/truth [%]			F_1	HS res
		incl.	PU200	$\mu < 20$	[%]	[μm]
A: max Clean/truth	0.98	73.83	67.73	84.22	72.09	9.70
D: most μ-robust	**0.97**	73.23	66.99	86.45	74.45	9.53
B: max HS selection	0.94	68.70				
C: max track F_1	0.92	65.84				
E: no cut	0	56.79				
AMVF		40.49	30.94	77.06	74.43	9.93

Test events, v6 peaks (12 November freeze). D: largest worst-case ratio, over five μ bins, of Clean/truth to its best value in the bin.

- Checkpoint: epoch 527, the lowest validation-monitor loss (0.255807) of 540 shard epochs; plateaued at the learning-rate floor.
- $t = 0.97$ is within 0.6 points of the maximum Clean/truth, gains 2.2 points at $\mu < 20$, 2.4 points of F_1 and 0.17 μm of HS resolution.
- $t = 0.99$ is worse than 0.98 in Clean/truth, F_1, HS selection and HS resolution.
- Chosen by Mattia, 12 November; also used at low pile-up.

`gnn_v2_freeze_2026-11-12.md Sec. 1, 5; G/options.md`

M4. The t Scan in μ Bins



All chain metrics versus t per μ bin, GNN validation events, s8 peaks.

M5. Behaviour of the GNN at Low μ

Seen at $\mu < 20$: Clean/truth peaks at $t = 0.90$–0.92 and falls at higher t; above $\mu = 50$ it rises up to 0.98.

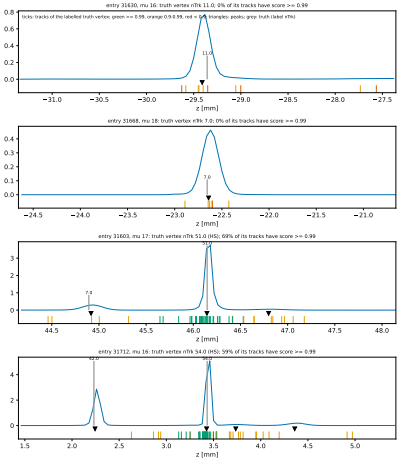
Why, measured:

- At $\mu < 10$, 97% of top-1 edges are correct (71% at μ 150–200). A higher t then removes mostly good tracks.
- The network is less confident on small vertices at low μ: 47% of correct tracks of 5–10-track vertices score below 0.99 at $\mu < 10$ (25% at μ 20–50, 41% at 150–200), on truth-candidate graphs too.
- $\mu < 20$ events hold about 2% of the tracks of a flat-μ sample, so they weigh little in the loss.
- v1 was worse (47% vs 33% of correct tracks below 0.99 at $\mu < 10$): the behaviour Mattia saw.
- Splits at low μ are HS satellites: 60% have the HS as dominant truth, 63% hold one track.

- Fake/reco stays at or below 0.4%; there is no failure mode.
- The satellite splits are weak density peaks next to the large HS peak, a larger share of the list when there are few vertices. They are not GNN errors.
- $t = 0.97$ is the compromise: within 0.6 points of the best Clean/truth at high μ, 2.2 points better than 0.98 at $\mu < 20$.

At the working point, μ 0–10: track F_1 TBD% (PV-Finder + GNN) vs TBD% (AMVF). No track-F_1 claim at low μ in the paper.

M5. Low-μ Event Displays



Validation events at $\mu \approx 17$: isolated 7- and 11-track vertices whose tracks all score 0.9–0.99, so the vertex vanishes at $t = 0.99$.

M6. CPU Timing against AMVF

Projected Athena CPU time (Xeon 4216, one thread), from speed-ups measured on sneezy and the note's Athena table:

$$T_{\text{chain}} = \frac{2189\ \text{ms}}{S} + 31\ \text{ms} + 182\ \text{ms}$$

Chain	S	T_{chain} [ms]
v6 padded (Athena today)	1	2401
s8 packed + GNN v2	TBD	TBD
s8 padded + GNN v2	TBD	TBD
AMVF (note figure, $\mu = 200$)		650 (600–750)

- With packed inference the chain is TBD× faster than AMVF on CPU. The worst case tried, TBD ms, is still below 600 ms.
- Packed inference needs a new Athena tensor builder: a track list plus an interval index.
- With today's padded tensor, s8 is comparable to AMVF, not faster.
- The GNN is now the largest step (182 ms in Athena).
- On GPU the full chain takes 76 ms (note).
- AMVF cannot run on sneezy: its time is read off the note's figure. The chain time is a projection, not an Athena measurement.

`H/cpu_timing/README.md; note sections/04b_timing.tex, tab:athena-timing-breakdown`

M7. Binning: Is the Network or the 0.04 mm Grid the Limit?

Professor's question: does the smaller Run 2 → HL-LHC gain of the AMVF/PV-Finder ratio mean that PV-Finder does not use the better HL-LHC tracks, so that finer bins are needed?

Test: the training target is what a perfect network would output. It goes through the same peak finder and the same metrics.

$\sigma_{\text{vtx}-\text{vtx}}$ [mm]	Run 2 MC	HL-LHC PU200
Target, 0.04 mm grid	0.203	0.145 (0.137*)
Target, 0.02 mm grid*		0.124
PV-Finder	0.410	0.218 (v6)
AMVF	0.789	0.305

*1-bin minimum width, so that every HS target spike is found.

- PV-Finder is 2.0 times its target in Run 2 and 1.5–1.7 times at HL-LHC: closer to its ideal at HL-LHC.
- A 0.02 mm grid improves the ideal by 10%, far from the gap to PV-Finder.
- The 0.04 mm grid carries the HS position to 0.77 μm; PV-Finder reaches 12.4 μm.
- Track-based ideal precision of pile-up vertices: 75.6 → 44.4 μm (1.7×). At the same peak setting PV-Finder $\sigma_{\text{vtx}-\text{vtx}}$ also improves 1.7× (0.410 → 0.241 mm).

The headroom is in the network, not in the grid. Finer bins are not needed.

M8. Hard-Scatter Selection with One Identical Rule

The Athena 24.0 rule (read from source):

- vertices sorted by $\sum p_{\mathrm{T}}^2$ of their linked tracks (`SumPt2Sorting`); the first is `PriVtx`;
- a track is linked if its compatibility with the vertex is $\chi^2 \leq 18.42$; no weights, no p_{T} cap;
- HS match: the vertex with the most HS tracks (`bestHardScatterMatch`).

Validation on AMVF (first 3,000 entries): on its stored lists our implementation agrees with the stored primary in 96.2% of events (94.8% at PU200); plain $\sum p_{\mathrm{T}}^2$ gives 86.8%.

PU200, reco and selected (0.1 mm)	[%]
PV-Finder + GNN, Athena rule	TBD ± TBD
AMVF, stored primary	92.92
AMVF, same rule, our code	89.11
AMVF, plain $\sum p_{\mathrm{T}}^2$ (old note)	78.80
PV-Finder + GNN − AMVF stored	TBD ± TBD pt

- Headline: GNN with the Athena rule against AMVF's own stored primary. Our code can only lose AMVF information, so this favours AMVF.
- It replaces the note's +3.4 points, which compared different rules.

`hs_selection_identical_rule_2026-11-11.md; src/gnn/evaluation/pv_selection.py`

M9. Hard-Scatter Resolution: Why PV-Finder + GNN Reaches AMVF

HS position (GNN v4 study)	[μm]
Histogram peak, local centroid	12.38
1D adaptive fit of z_0, GNN tracks	11.99
3D adaptive fit, all tracks within 1 mm	9.28
3D adaptive fit, AMVF's own HS tracks	9.50
3D adaptive fit, true HS tracks	9.09
AMVF stored vertex	10.25
Final, 3D fit, GNN tracks, $t = 0.97$	TBD $\pm$ TBD
Final, AMVF	10.24 ± 0.27

- The missing piece was the transverse vertex position. ITk tracks reach $|\eta| = 4$ (median $|\cot\theta| = 3.2$), so a 10–15 μm offset from the beam axis moves z_0 by tens to hundreds of μm.
- The fit: (x, y, z) with a beam-spot constraint, (d_0, z_0) covariance, adaptive weights with annealing, seeded at the histogram peak.
- GNN tracks do as well as the true HS tracks; on AMVF's own tracks the same fit beats AMVF's stored position. The fit model, not the association, made the difference.
- In the note: say only how it is computed.

`itk_hs_resolution_2026-11-09.md (2026-11-11 sections); hs_position_estimators.adaptive_beamline_fit`

M10. Resolution versus μ: the 68% Residual

Choice: the resolution versus μ in the paper is the 68% residual $|z_{\text{reco}} - z_{\text{true}}|$. $\sigma_{\text{vtx}-\text{vtx}}$ is quoted at PU200 only, or per bin for $\mu \geq 100$.

Why:

- Both algorithms make more close vertex pairs than truth has: close-pair rate over flat expectation 1.61 (PV-Finder), 1.60 (AMVF), 1.01 (truth) at $\mu < 20$. The hard-scatter neighbourhood is the largest source.
- This excess is about constant per event while random pairs grow as μ^2, so it dominates at low μ. Below $\mu = 100$ the sigmoid fit measures a notch inside it, not the separation (χ^2/ndf 4–57).
- $\sigma_{\text{vtx}-\text{vtx}}$ is not flat in μ for either algorithm, so a single all-μ value is wrong.
- The 68% residual is defined at every μ; the two algorithms agree within 2 μm in every bin.

`resolution_vs_mu_2026-11-11.md; H/resolution_vs_mu`

Figure not produced yet

`outputs/11_12_2026_output/hllhc_final/resolution_vs_mu/resolution_vs_mu.{pdf,png}`

image `resolution_vs_mu`; rebuild with build.sh when it exists

68% residual (top) and $\sigma_{\text{vtx}-\text{vtx}}$ (bottom; open markers below $\mu = 100$), s8.

6

Results

Every paper figure and number; PV-Finder and AMVF on the same events

HL-LHC before the GNN: Table 3 at PU200

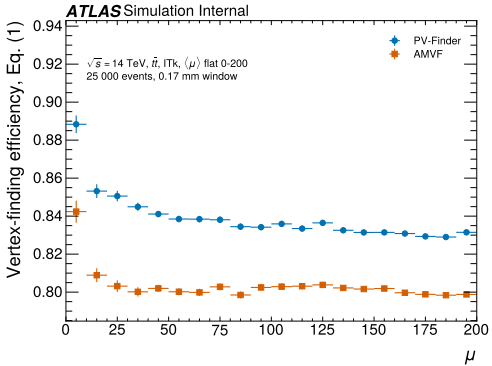
PAPER MAIN r16443, 1,920 events, $\langle\mu\rangle = 192.5$, 213,738 truth vertices ($n_{\text{trk}} \geq 2$), window 0.17 mm

	PV-Finder (s8)	AMVF	PVF − AMVF
Efficiency, Eq. (1) [%]	TBD ± TBD	79.85	TBD ± TBD pt
One-to-one efficiency [%]	TBD	TBD	
Fake / reco [%]	TBD	20.08	
Fakes / event	TBD	19.65	
Splits / event	TBD	0.184	
Reco vertices / event	TBD	97.9	(truth 111.3)
$\sigma_{\text{vtx−vtx}}$ [mm]	TBD ± TBD	0.305 ± 0.006	
68% residual [μm]	TBD	56.45	
HS reconstruction, 0.1 mm [%]	TBD	99.79	
Efficiency on r16638 (1,888 events) [%]	TBD	80.13	

HLDOC; H1 = H/h1_vertex_finding/h1_summary.json (table3.tex)

HL-LHC before the GNN: Efficiency and Fakes versus μ

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Vertex-finding efficiency, Eq. (1), 0.17 mm, r16443 entries 0–24,999.

Figure not produced yet

`outputs/11_12_2026_output/hllhc_final/h1_vertex_finding/plots/fake_fraction_vs_mu_r16443.{pdf,png}`

image `fake_fraction_vs_mu`; rebuild with build.sh when it exists

Fake fraction: reco vertices without truth within 0.17 mm.

HL-LHC before the GNN: Fakes and Yield versus μ

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Figure not produced yet

`outputs/11_12_2026_output/hllhc_final/h1_vertex_finding/plots/fakes_per_event_vs_mu_r16443.{pdf,png}`

image `fakes_per_event_vs_mu`; rebuild with build.sh when it exists

Fake vertices per event.

Figure not produced yet

`outputs/11_12_2026_output/hllhc_final/h1_vertex_finding/plots/fig6_vertices_vs_mu_r16443.{pdf,png}`

image `yield_vs_mu`; rebuild with build.sh when it exists

Reconstructed vertices per event, with truth (draft Fig. 6).

HL-LHC before the GNN: Vertex Separation and Position at PU200

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Figure not produced yet

`outputs/11_12_2026_output/hllhc_final/h1_vertex_finding/plots/paper_figures_r16443/fig4_hllhc_deltaz.{pdf,png}`

image `deltaz_pu200`; rebuild with build.sh when it exists

Δz between reconstructed vertices with the $\sigma_{\text{vtx}-\text{vtx}}$ fit (draft Fig. 7a).

Figure not produced yet

`outputs/11_12_2026_output/hllhc_final/h1_vertex_finding/plots/fig7b_residuals_r16443.{pdf,png}`

image `residuals_pu200`; rebuild with build.sh when it exists

Single-vertex residual $z_{\text{reco}} - z_{\text{true}}$, one-to-one within 1 mm (draft Fig. 7b).

HL-LHC before the GNN: Resolution versus μ

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Figure not produced yet

`outputs/11_12_2026_output/hllhc_final/resolution_vs_mu/resolution_vs_mu_p68.{pdf,png}`

image `p68_vs_mu`; rebuild with build.sh when it exists

68% residual versus μ: the paper's resolution versus pile-up. 44,999 events, r16443 and r16638.

	PV-Finder	AMVF
68% residual, μ 0–10 [μm]	TBD	32.1
68% residual, μ 180–200 [μm]	TBD	56.5
$\sigma_{\mathrm{vtx-vtx}}$, μ 100–120 [μm]	TBD	275
$\sigma_{\mathrm{vtx-vtx}}$, μ 180–200 [μm]	TBD	304
AMVF / PVF $\sigma_{\mathrm{vtx-vtx}}$, $\mu \geq 100$	TBD	

$\sigma_{\mathrm{vtx-vtx}}$ per bin only for $\mu \geq 100$ (M10). The two-panel version is on the M10 slide.

HL-LHC before the GNN: Window Justification and Scan

Figure not produced yet

`outputs/11_12_2026_output/hllhc_final/h1_vertex_finding/plots/window_justification_r16443.{pdf,png}`

image `window_justification`; rebuild with build.sh when it exists

PAPER MAIN $|z_{\mathrm{reco}} - z_{\mathrm{true}}|$ with the 68% points and $3\times$ the larger one: the 0.17 mm window.

Figure not produced yet

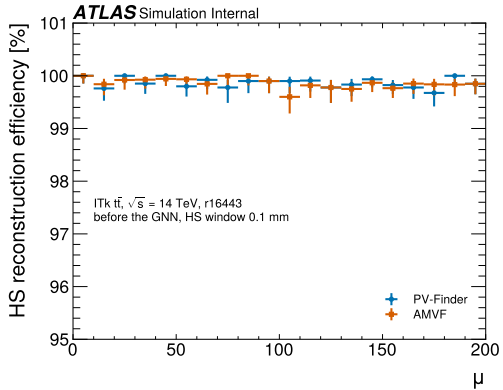
`outputs/11_12_2026_output/hllhc_final/h1_vertex_finding/plots/window_scan_r16443.{pdf,png}`

image `window_scan`; rebuild with build.sh when it exists

BACKUP / NOTE PV-Finder minus AMVF efficiency versus window.

HL-LHC before the GNN: Hard-Scatter Reconstruction

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Fraction of events with a reconstructed vertex within 0.1 mm of the true hard scatter, versus μ.

PU200: PV-Finder TBD%, AMVF 99.79%.

HL-LHC before the GNN: Backup

BACKUP / NOTE

Figure not produced yet

`outputs/11_12_2026_output/hllhc_final/h1_vertex_finding/plots/eff_one_to_one_vs_mu_r16443.{pdf,png}`

image `eff_one_to_one_vs_mu`; rebuild with build.sh when it exists

One-to-one efficiency versus μ.

Figure not produced yet

`outputs/11_12_2026_output/hllhc_final/h1_vertex_finding/plots/splits_per_event_vs_mu_r16443.{pdf,png}`

image `splits_per_event_vs_mu`; rebuild with build.sh when it exists

Split vertices per event versus μ.

HL-LHC before the GNN: Positional Categories (Backup)

BACKUP / NOTE

Figure not produced yet

`outputs/11_12_2026_output/hllhc_final/h1_vertex_finding/plots/paper_figures_r16443/fig3a_hllhc_categories_pu200.{pdf,png}`

image `pos_categories_pu200`; rebuild with build.sh when it exists

Positional categories at PU200, both algorithms.

Figure not produced yet

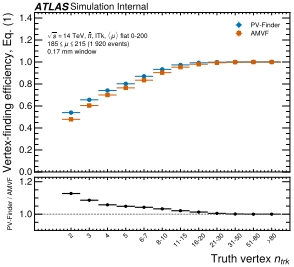
`outputs/11_12_2026_output/hllhc_final/h1_vertex_finding/plots/fig6_categories_vs_mu_amvf_r16443.{pdf,png}`

image `amvf_pos_categories_vs_mu`; rebuild with build.sh when it exists

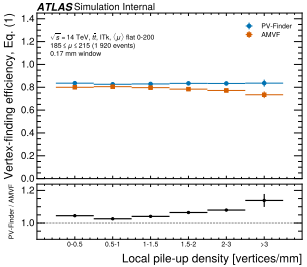
AMVF positional categories versus μ.

HL-LHC before the GNN: Efficiency vs n_{trk}, Density, Separation (Backup)

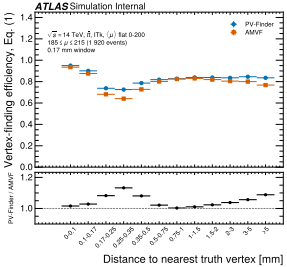
BACKUP / NOTE



versus truth n_{trk}, PU200.



versus local density, PU200.



versus Δz to the nearest truth vertex, PU200.

HL-LHC after the GNN: Table 4 at PU200

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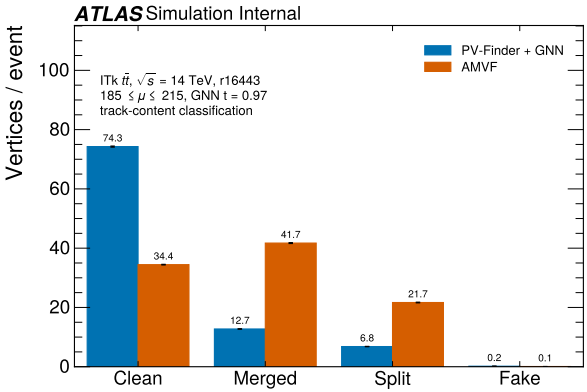
s8 + GNN v2 ($t = 0.97$) vs AMVF TTVA links, 1,920 events, vertices with ≥ 1 track

	PV-Finder + GNN	AMVF
Clean / Merged / Split / Fake per event	TBD / TBD / TBD / TBD	34.4 / 41.7 / 21.7 / 0.07
Clean / Merged / Split / Fake per reco [%]	TBD / TBD / TBD / TBD	35.2 / 42.6 / 22.1 / 0.07
Clean / truth [%]	TBD $\pm$ TBD	30.94
HS reconstructed and selected [%]	TBD $\pm$ TBD	92.92 (stored primary)
HS longitudinal resolution [μm]	TBD $\pm$ TBD	10.24 ± 0.27
Clean-HS fraction [%]	TBD	92.5
HS-track fraction [%]	TBD	74.1
Track efficiency / purity / F_1 [%]	TBD / TBD / TBD	TBD / TBD / TBD
Wrong-assignment rate [%]	TBD	TBD
All μ (25,000 events): Clean/truth [%]	TBD	40.49
All μ: HS reconstructed and selected [%]	TBD	95.14

HLDOC; CH, CHI, SUP

HL-LHC after the GNN: Pile-up Vertex Quality at PU200

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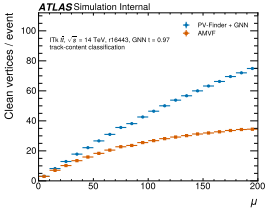


Clean, Merged, Split and Fake vertices per event at PU200, track-content classification (70% rule), PV-Finder + GNN tracks vs AMVF TTVA links.

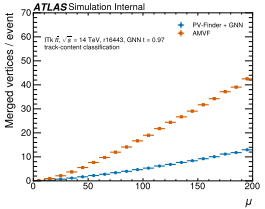
Clean per event: TBD vs 34.4.

HL-LHC after the GNN: Vertex Categories versus μ

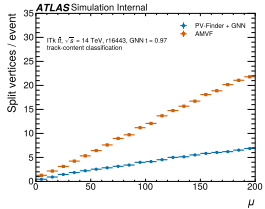
PAPER MAIN



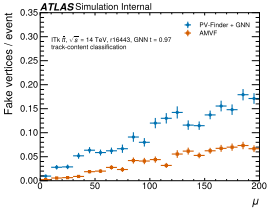
Clean / event



Merged / event



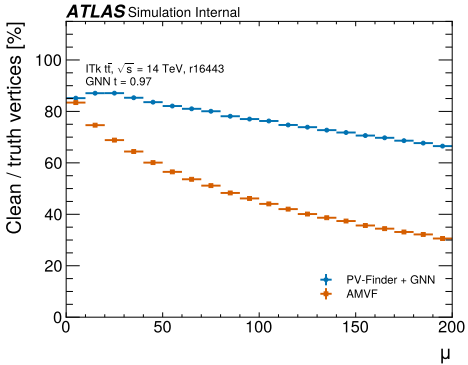
Split / event



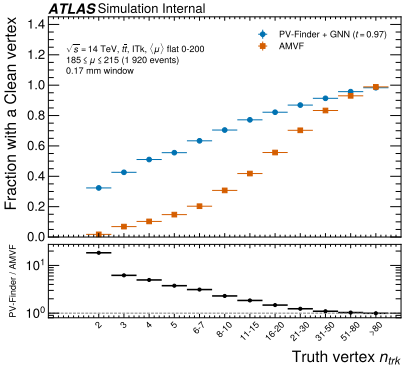
Fake / event

HL-LHC after the GNN: Clean/truth and Efficiency versus n_{trk}

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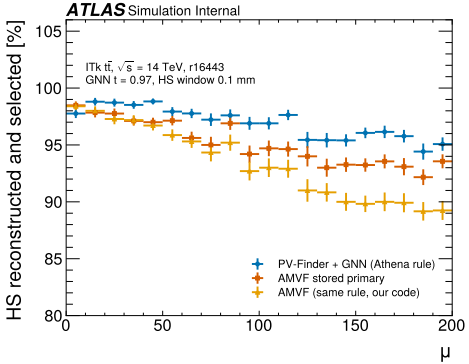
Clean vertices over truth vertices versus μ.



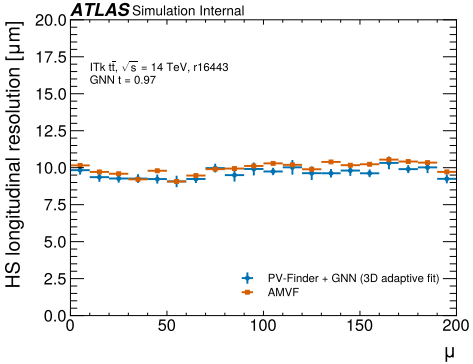
Efficiency after the GNN versus truth n_{trk}, PU200 (PUB-note style).

HL-LHC after the GNN: Hard-Scatter Selection and Resolution

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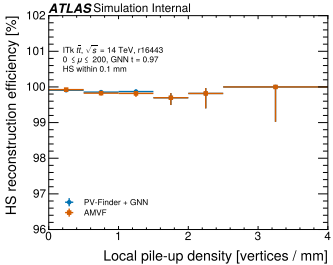
HS reconstructed and selected (0.1 mm), Athena rule for PV-Finder + GNN; AMVF stored primary and same rule.

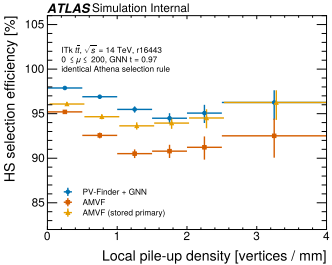


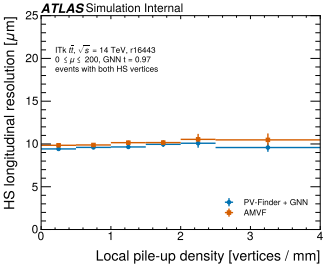
HS longitudinal resolution (iterative 3-RMS), 3D adaptive fit on GNN tracks vs AMVF.

HL-LHC after the GNN: Hard Scatter versus Local Pile-up Density

PAPER MAIN



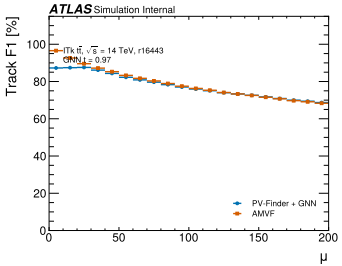




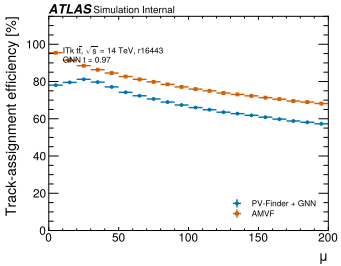
HS reconstruction HS selection HS resolution

Density: truth vertices within ± 2 mm of the true HS, HS excluded, per mm (ITk). All μ, r16443 entries 0–24,999.

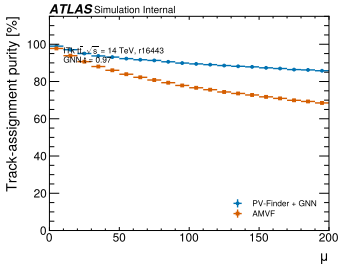
HL-LHC after the GNN: Track Association



PAPER MAIN Track F_1



BACKUP / NOTE Track efficiency

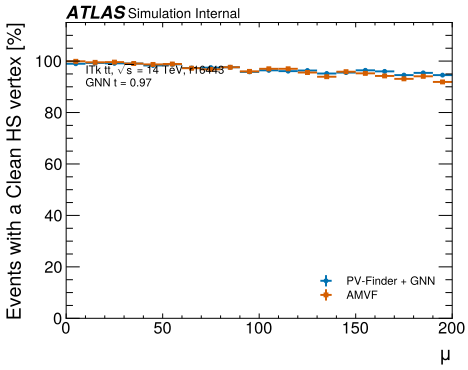


BACKUP / NOTE Track purity

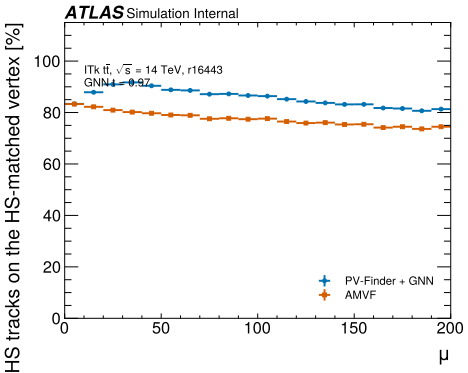
At PU200 the GNN trades efficiency (TBD vs TBD%) for purity (TBD vs TBD%); F_1 TBD vs TBD%. At low μ the GNN F_1 is below AMVF (M5).

HL-LHC after the GNN: Hard-Scatter Tracks (Backup)

BACKUP / NOTE



Clean-HS fraction versus μ.



HS tracks on the HS-matched vertex over all HS tracks.

HL-LHC after the GNN: HS Resolution vs n_{trk}, Density Definition (Backup)

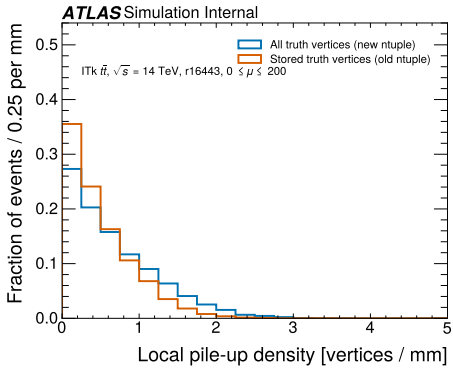
BACKUP / NOTE

Figure not produced yet

`outputs/11_12_2026_output/hllhc_final/chain/plots/hs_resolution_vs_ntrk.{pdf,png}`

image `hs_res_vs_ntrk`; rebuild with build.sh when it exists

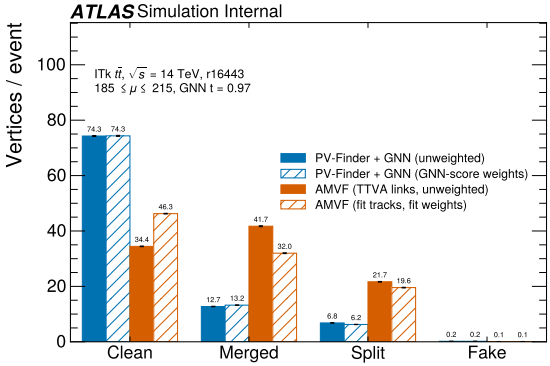
HS resolution versus HS truth n_{trk}, PU200.



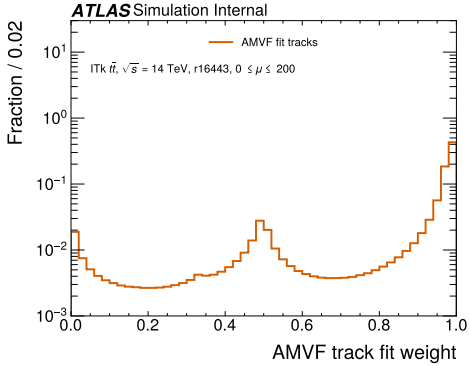
Local density: ITk definition (all truth vertices) and the stored-truth one.

HL-LHC after the GNN: Classification Cross-checks (Backup)

BACKUP / NOTE



Categories per event at PU200: unweighted (headline), GNN-score weights, AMVF fit tracks with fit weights.

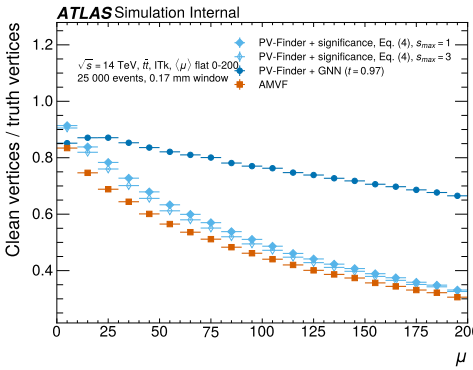


AMVF fit-weight distribution.

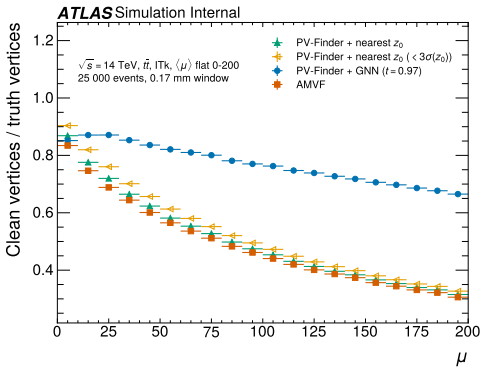
With its own fit weights AMVF has TBD Clean per event; PV-Finder + GNN leads by TBD× instead of TBD×.

HL-LHC after the GNN: Association Baselines (Backup)

BACKUP / NOTE



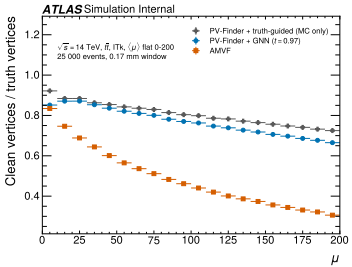
Eq. (4) significance association, Clean/truth versus μ.



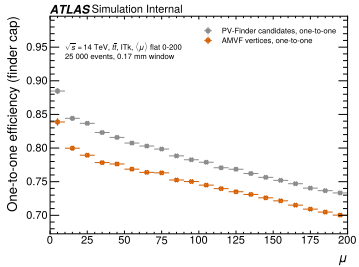
Nearest z_0 and $3\sigma(z_0)$ baselines.

HL-LHC after the GNN: Bounds and Stages (Backup)

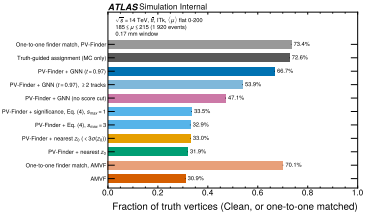
BACKUP / NOTE



Truth-guided association on the same peaks.



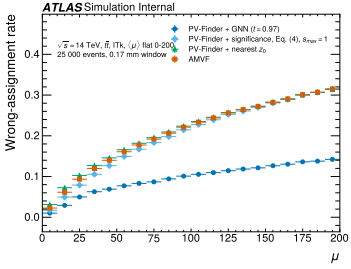
Finder cap (one to one, 0.17 mm).



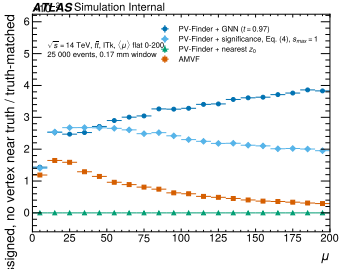
Clean yield by stage, PU200.

HL-LHC after the GNN: Unassigned and Wrong Tracks (Backup)

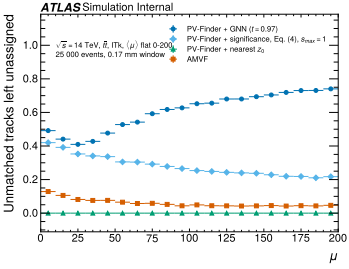
BACKUP / NOTE



Wrong-assignment rate (1− purity).



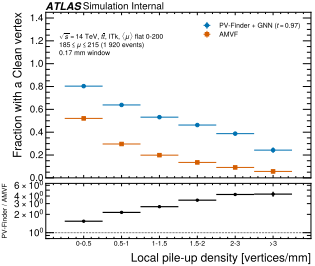
Labelled tracks left unassigned with no candidate within 0.17 mm.



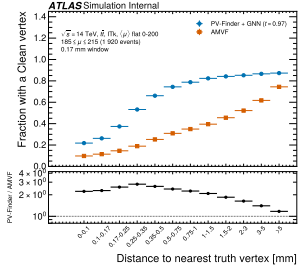
Unlabelled tracks correctly left unassigned.

HL-LHC after the GNN: Density, Separation, Peak Width (Backup)

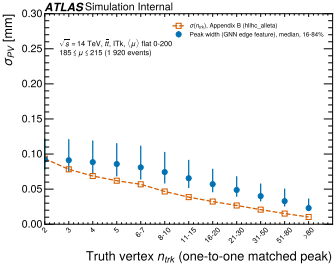
BACKUP / NOTE



Efficiency after the GNN versus local density.



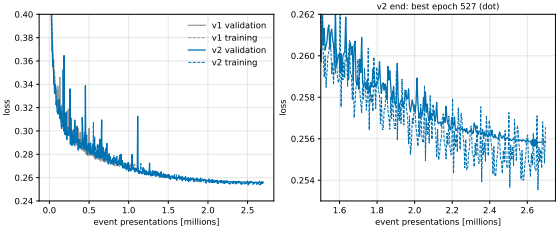
versus Δz to the nearest truth vertex.



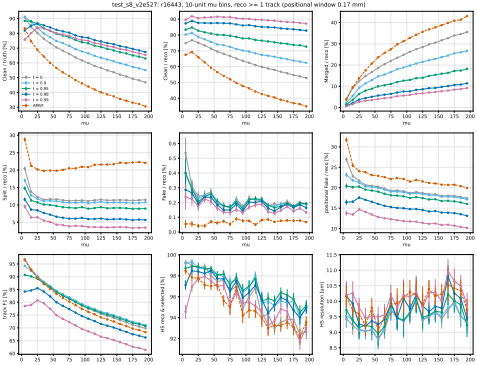
Peak width (σ_{PV}) vs the n_{trk} parameterisation.

GNN v2: Learning Curve and Metrics versus μ (Backup)

BACKUP / NOTE



Validation-monitor loss, v1 and v2.



All chain metrics versus μ for several t, s8 peaks, test events.

Low Pile-up: Vertex Finding (Table 6)

PAPER MAIN

PVF-e400, integral 0.50, window 0.17 mm

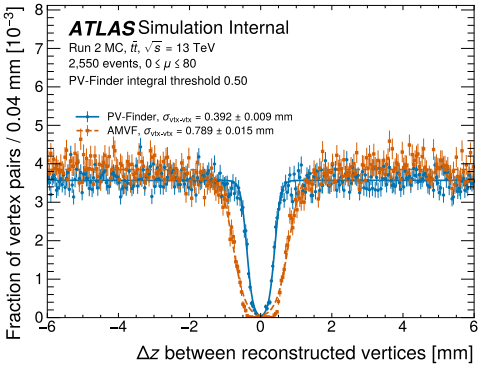
Run 2 MC, 2,550 events, 72,189 truth vertices	PV-Finder	AMVF	PVF − AMVF
Efficiency [%]	TBD ± TBD	TBD	TBD pt
Fake / reco [%]	TBD	TBD	
Fakes / event	TBD	TBD	
$\sigma_{\mathrm{vtx-vtx}}$ [mm]	TBD ± TBD	TBD ± TBD	
68% residual [μm]	TBD	TBD	
HS reconstruction, 0.1 mm [%]	TBD	TBD	

Collision data, AMVF as reference	Run 2 data	Run 3 data	Run 2 MC, same convention
AMVF-vertex recovery [%]	TBD	TBD	TBD
PVF vertices without AMVF vertex / event	TBD	TBD	
$\sigma_{\mathrm{vtx-vtx}}$ PV-Finder / AMVF [mm]	TBD / TBD	TBD / TBD	

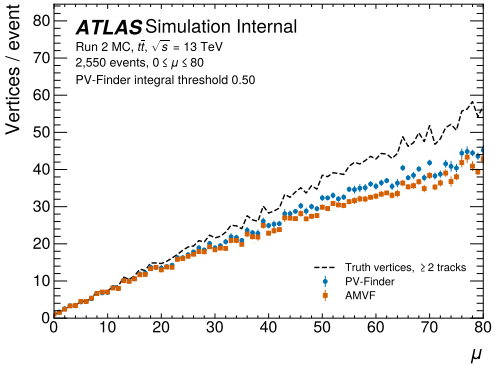
LPDOC R1, R3; VF

Low Pile-up: Run 2 MC Vertex Finding (Fig. 8)

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(a) Δz between vertices, PV-Finder and AMVF.



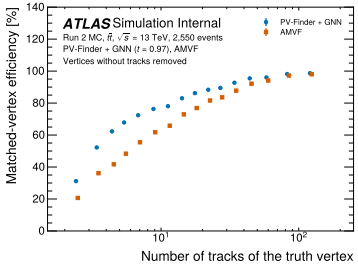
(b) Vertices per event versus μ.

Low Pile-up: Run 2 MC after the GNN (Table 5, Fig. 9)

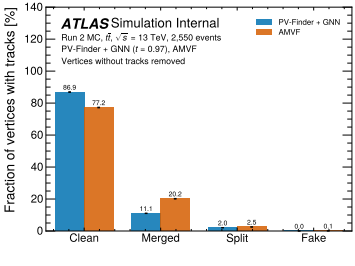
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[%]	PVF + GNN	AMVF
Clean	TBD	TBD
Merged	TBD	TBD
Split	TBD	TBD
Fake	TBD	TBD
Clean/truth	TBD	TBD
Track F_1	TBD	TBD
HS ident.	TBD	TBD

$t = 0.97$; per reco vertex with ≥ 1 track; TBD empty candidates dropped.



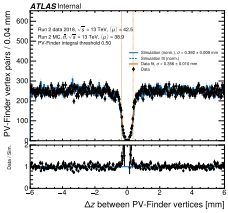
(a) Matched-vertex efficiency vs truth n_{trk}.



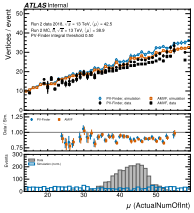
(b) Categories.

Low Pile-up: Collision Data (Fig. 10)

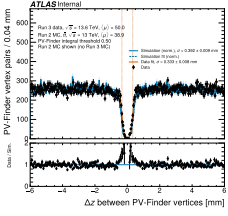
PAPER MAIN



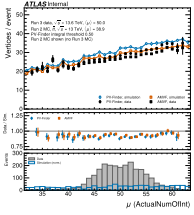
(a) Run 2 data Δz, with Run 2 MC



(b) Run 2 data vertices vs μ



(c) Run 3 data Δz, with Run 2 MC



(d) Run 3 data vertices vs μ

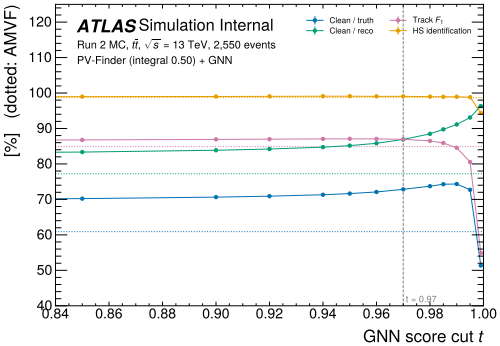
Low Pile-up: Run 3 Data Association and the t Scan

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Run 3 data, 2,500 events, $t = 0.97$	[%]
Tracks assigned by the GNN	TBD
Agreement with AMVF (1 mm)	TBD
Agreement with AMVF (0.17 mm)	TBD
Leading-vertex agreement	TBD

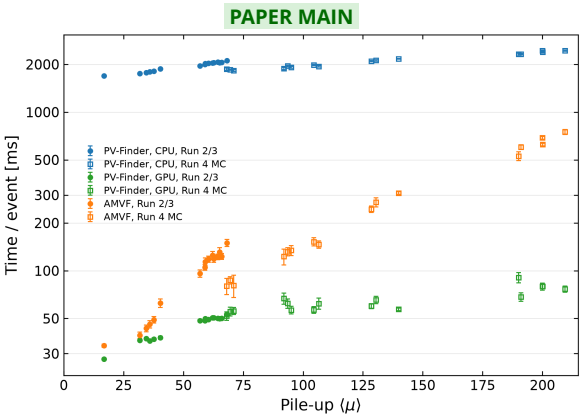
Same events and peaks as the vertex finding.
Agreement, not efficiency.

BACKUP / NOTE



Run 2 MC metrics versus t.

Timing: CPU and GPU



Athena timing versus pile-up: AMVF on CPU, PV-Finder chain on GPU (note figure).

$\mu = 200$, per event	[ms]
PV-Finder chain, GPU (Athena)	76
PV-Finder chain, CPU (Athena, v6)	2401
s8 chain, CPU, projected	TBD
AMVF, CPU (note figure)	650

The CPU claim in the paper needs packed inference in Athena (M6). The GPU claim stands as measured.

7

Proposed Internal-Note Edits

Old text and new [MM: ...] text, with page and line

Note Edits: List

Internal note, GitLab main. Changes in red as [MM: ...] on a branch with a merge request.

1. The 0.17 mm window with its justification plot; the hard scatter also at 0.1 mm.
2. Split-inclusive efficiency: one sentence.
3. HS longitudinal resolution from the 3D adaptive fit: one paragraph and one figure.
4. Timing claim restricted to GPU.
5. Simulation on the Fig. 6.2 panels, and the AMVF-reference bias.
6. z_0 baselines in Fig. 7.2.
7. AMVF HS selection 92.92% and the HS numbers.
8. The AMVF-weights paragraph.
9. The frozen configuration: s8, GNN v2 epoch 527, $t = 0.97$; PVF-e400 at low pile-up.

Note Edit 1: Matching Window

Note page, line: TBD

Section: TBD

Old text

TBD

New text, in red [MM: ...]

TBD

Note Edit 2: Split-Inclusive Efficiency

Note page, line: TBD

Section: TBD

Old text

TBD

New text, in red [MM: ...]

TBD

Note Edit 3: Hard-Scatter Longitudinal Resolution

Note page, line: TBD

Section: TBD

Old text

TBD

New text, in red [MM: ...]

TBD

Note Edit 4: Timing Claim

Note page, line: TBD

Section: TBD

Old text

TBD

New text, in red [MM: ...]

TBD

Note Edit 5: Fig. 6.2 and the AMVF Reference

Note page, line: TBD

Section: TBD

Old text

TBD

New text, in red [MM: ...]

TBD

Note Edit 6: z_0 Baselines in Fig. 7.2

Note page, line: TBD

Section: TBD

Old text

TBD

New text, in red [MM: ...]

TBD

Note Edit 7: AMVF Hard-Scatter Selection

Note page, line: TBD

Section: TBD

Old text

TBD

New text, in red [MM: ...]

TBD

Note Edit 8: AMVF Weights

Note page, line: TBD

Section: TBD

Old text

TBD

New text, in red [MM: ...]

TBD

Note Edit 9: Frozen Configuration

Note page, line: TBD

Section: TBD

Old text

TBD

New text, in red [MM: ...]

TBD